1 Appendix

1.1 Theoretical Guarantees

First we present lemmas, which will support our theorems. We note that the this analysis goes more or less along the same lines of [1].

Let us define the event $\mathcal{F}_p(t) = \left\{ \exists i \in [N],\ j \in [K] : |\hat{\mu}_j^{(i)} - \mu_j^{(i)}| > \sqrt{\frac{2\log t}{T_j^{(i)}}} \right\}$, a player's bad event that some preference of the arms are not estimated by the players correctly at time t. We present the following Lemma 5.1, Lemma 5.4 ,and Lemma 5.5 of [1]

Lemma 1.1.

$$\mathbb{E}\left[\sum_{t=1}^{T} 1\{\mathcal{F}_p(t)\}\right] \leq \frac{NK\pi^2}{3} \tag{1}$$

Lemma 1.2. *Conditional on $\mathcal{F}_p(t)^{\complement}$, $\mathrm{UCB}_j^{(i)}(t) < \mathrm{LCB}_{j'}^{(i)}(t)$ implies $\mu_j^{(i)} < \mu_{j'}^{(i)}$.*

Lemma 1.3. *In round t, let $T^{(i)}(t) = \min_{j\in[K]} T_j^{(i)}(t)$ and $\bar{T}^{(i)} = 32\log T/(\Delta^{(i)})^2$ where $\Delta^{(i)} = \min_{j\neq j'} \Delta_{j,j'}^{(i)} = \min_{j\neq j'} |\mu_j^{(i)} - \mu_{j'}^{(i)}|$. Conditional on $\mathcal{F}_p(t)^{\complement}$, if $T^{(i)} > \bar{T}^{(i)}$ we have $\mathrm{UCB}_j^{(i)}(t) < \mathrm{LCB}_{j'}^{(i)}(t)$ for any $j, j' \in [K]$ with $\mu_j^{(i)} < \mu_{j'}^{(i)}$.*

Lemma 1.3 can similarly be written for arms. Following we extend the Lemma 5.3 from [1] to two-sided learning

Define the event $\mathcal{F}_a(t) = \left\{ \exists i \in [N],\ j \in [K] : |\hat{\eta}_i^{(j)} - \eta_i^{(j)}| > \sqrt{\frac{2\log t}{T_i^{(j)}}} \right\}$, an arm's bad event that some preference of the players are not estimated by the arms correctly at time t.

Lemma 1.4. *Conditional on $\cap_{t=1}^{T}(\mathcal{F}_p(t) \cup \mathcal{F}_a(t))^{\complement}$, after epoch $l_{\max}$, the player-optimal regret and arm-pessimal regret for the period $b^l T_0 - 2^l T_0 - K^2$ in epoch $l > l_{\max}$ is zero where*

$$l_{\max} = \min\left\{ l : \sum_{l'=1}^{l} 2^{l'} T_0 \geq 32K\log T/\Delta^2 \right\}. \tag{2}$$

Proof. Since in each exploration period all players propose to distinct arms using a round-robin fashion, no collision occurs and all players are accepted at each round of the exploration period in each epoch. Thus at the end of epoch $l_{\max}$ it holds that $T_j^{(i)} \geq 32\log T/(\Delta^{(i)})^2$ and $T_i^{(j)} \geq 32\log T/(\Delta'^{(j)})^2$ for any $i \in [N]$ and $j \in [K]$ since $\Delta = \min_{i\in[N], j\in[K]}\{\Delta^{(i)}, \Delta'^{(j)}\}$.
Now according to Lemma 1.3, when $T_j^{(i)} \geq 32\log T/(\Delta^{(i)})^2$ for any arm a_j, player p_i finds a permutation $\sigma^{(i)}$ over arms such that $\mathrm{LCB}_{\sigma_k^{(i)}}^{(i)} < \mathrm{UCB}_{\sigma_{k+1}^{(i)}}^{(i)}$ for any $k \in [K-1]$. Similarly, for any player p_i, arm a_j finds a permutation $\sigma'^{(j)}$ over players such that $\mathrm{LCB}_{\sigma_k^{(j)}}^{(j)} < \mathrm{UCB}_{\sigma_{k+1}^{(j)}}^{(j)}$ for any $k \in [N-1]$.
This implies that after $l > l_{\max}$, all players and arms finds a permutation of arms and players resp. with disjoint confidence intervals. Since, Gale-Shapley algorithm will last at most K^2 steps, we will have zero regret in the period $b^l T_0 - 2^l T_0 - K^2$ conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t) \cup \mathcal{F}_a(t))^{\complement}$. □

1.1.1 Proof of Theorem 1

Proof of Theorem 1. From Lemma 1.3 after $t_{\exp} = 32K\log T/\Delta^2$, conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t))^{\complement}$ every player p_i finds *true* preference ranking over arms and similarly, conditioned on the event $\cap_{t=1}^{T}(\mathcal{F}_p(t))^{\complement}$ every arm a_j finds correct preference ranking over players, thus the ith and $(N+j)^{\text{th}}$ entry of blackboard is 1. Thus everyone enters Gale-Shapley rounds after $t_{\exp}$

number of exploration rounds. Thus the optimal stable regret for player p_i is

$$\begin{aligned}
\overline{RP}_i(T) &= \mathbb{E}\left[\sum_{t=1}^{T} \mu^{(i)}_{\bar{m}_i} - X^{(i)}(t)\right] \\
&\leq \mathbb{E}\left[\sum_{t=1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i\}\Delta^{(i)}_{\max}\right] \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\mathcal{F}_p(t)\} + 1\{\mathcal{F}_a(t)\}\right]\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \frac{32K\log T}{\Delta^2}\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max}
\end{aligned}$$

□

1.1.2 Proof of Theorem 2

We will find an equation of $l_{\max}$, from the definition we have and use the fact that $b > 2$ to get

$$\begin{aligned}
l_{\max} &= \left\lceil \log\left(\frac{32K\log T}{T_0(\Delta^{(i)})^2} + 2\right) - 1\right\rceil = \left\lceil \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right)\right\rceil \\
\implies \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right) &\leq l_{\max} < \log\left(\frac{16K\log T}{T_0(\Delta^{(i)})^2} + 1\right) + 1 = \log\left(\frac{32K\log T}{T_0(\Delta^{(i)})^2} + 2\right)
\end{aligned}$$

Define $\tilde{l}$, the epoch number required to complete T number of interaction rounds. We have from the definition

$$\begin{aligned}
\sum_{l=1}^{\tilde{l}} b^l T_0 &= T - N \\
\frac{b^{\tilde{l}+1} - b}{b-1} T_0 &= T - N \\
b^{\tilde{l}+1} &= \frac{(T-N)}{T_0}(b-1) + b \\
\tilde{l} &= \log_b\left(\frac{b-1}{b}\frac{(T-N)}{T_0} + 1\right) < \log\left(\frac{T-N}{T_0} + 1\right)^{\log_b 2} = \log\left(\frac{T-N}{T_0} + 1\right)^{\gamma}
\end{aligned}$$

Proof of Theorem 2. The optimal stable regret for player p_j is the sum of regret before and after $l_{\max}$.

$$\begin{aligned}
\overline{RP}_i(T) &= \mathbb{E}\left[\sum_{t=1}^{T} \mu^{(i)}_{\bar{m}_i} - X^{(i)}(t)\right] \\
&\leq \mathbb{E}\left[\sum_{t=1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i\}\Delta^{(i)}_{\max}\right] \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\mathcal{F}_p(t)\} + 1\{\mathcal{F}_a(t)\}\right]\Delta^{(i)}_{\max} \\
&\leq N\Delta^{(i)}_{\max} + \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right]\Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3}\Delta^{(i)}_{\max}
\end{aligned}$$

Let $T_1 = \sum_{l=1}^{l_{\max}} b^l T_0$ and $T_2 = \sum_{l=l_{\max}+1}^{\tilde{l}} b^l T_0$ be the time-period before and after epoch $\tilde{l}$ with $T - N = T_1 + T_2$. Thus we have

$$R(T) = \mathbb{E}\left[\sum_{N+1}^{T} 1\{\bar{A}(t) \neq \bar{m}_i, \mathcal{F}_p(t)^{\complement}\}\right] \Delta_{\max}^{(i)} = \mathbb{E}\left[\sum_{l=1}^{l_{\max}} b^l T_0\right] \Delta_{\max}^{(i)} + \mathbb{E}\left[\sum_{l=l_{\max}+1}^{\tilde{l}} 2^l T_0 + K^2\right] \Delta_{\max}^{(i)}$$

since the regret before epoch $l_{\max}$ i.e. for $l \leq l_{\max}$ is $b^l T_0$ and after $l_{\max}$ we have non-zero regret only from exploration and Gale-Shapley rounds due to Lemma 1.4. Thus the total regret is

$$\begin{aligned}
R(T) &= \mathbb{E}\left[\sum_{l=1}^{l_{\max}} b^l T_0\right] \Delta_{\max}^{(i)} + \mathbb{E}\left[\sum_{l=l_{\max}+1}^{\tilde{l}} 2^l T_0 + K^2\right] \Delta_{\max}^{(i)} \\
&\leq T_0 b^{l_{\max}+1} \Delta_{\max}^{(i)} + T_0(2^{\tilde{l}+1} - 2^{l_{\max}+1}) \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} \\
&= 2^{\tilde{l}+1} T_0 \Delta_{\max}^{(i)} + (b^{l_{\max}+1} - 2^{l_{\max}+1}) T_0 \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} \\
&\leq 2^{\tilde{l}+1} T_0 \Delta_{\max}^{(i)} + (l_{\max}+1) b^{l_{\max}} (b-2) T_0 \Delta_{\max}^{(i)} + K^2 \tilde{l} \Delta_{\max}^{(i)} && \text{(convexity of } x^{l_{\max}+1} \text{ for } l_{\max} \geq 1) \\
&\leq 2\left(\frac{T}{T_0}\right)^{\gamma} T_0 \Delta_{\max}^{(i)} + \log\left(\frac{64K \log T}{T_0 (\Delta^{(i)})^2} + 4\right)\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right)^{1/\gamma} (2^{1/\gamma} - 2) T_0 \Delta_{\max}^{(i)} + K^2 \gamma \log\left(\frac{T}{T_0}\right) \Delta_{\max}^{(i)}
\end{aligned}$$

We use the following inequality of convex functions

$$f(x) - f(y) \leq \nabla f(x)(x - y).$$

Thus we have in total

$$\begin{aligned}
\overline{RP}_i(T) \leq N \Delta_{\max}^{(i)} &+ 2\left(\frac{T}{T_0}\right)^{\gamma} T_0 \Delta_{\max}^{(i)} + \log\left(\frac{64K \log T}{T_0 (\Delta^{(i)})^2} + 4\right)\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right)^{1/\gamma} (2^{1/\gamma} - 2) T_0 \Delta_{\max}^{(i)} \\
&+ \ \gamma K^2 \log\left(\frac{T}{T_0}\right) \Delta_{\max}^{(i)} + \frac{2NK\pi^2}{3} \Delta_{\max}^{(i)}
\end{aligned}$$

For T_2 to be defined correctly we need that

$$\begin{aligned}
\log\left(\frac{T-N}{T_0} + 1\right)^{\gamma} > \tilde{l} \geq l_{\max} + 1 &\geq \log\left(\frac{32K \log T}{T_0 (\Delta^{(i)})^2} + 2\right) \\
\implies \left(\frac{T-N}{T_0}\right)^{\gamma} &\gtrsim \frac{32K \log T}{T_0 (\Delta^{(i)})^2} \\
T_0 &\gtrsim \left(\frac{32K \log T}{(\Delta^{(i)})^2 (T-N)^{\gamma}}\right)^{\frac{1}{1-\gamma}} \quad \forall i \\
\implies T_0 &\gtrsim \left(\frac{32K \log T}{\Delta^2 (T-N)^{\gamma}}\right)^{\frac{1}{1-\gamma}}
\end{aligned}$$

□

1.2 Arm's Learning

1.2.1 Known Gap

Algorithm 1: Expore-then-Gale-Shapley (view of arm a_j) with Known Gap Δ

Input : Exploration rounds $t_{\text{exp}} = \left\lceil \frac{32K \log T}{\Delta^2} \right\rceil$

`// Learning Preferences`

for $t = 1, \ldots, t_{\text{exp}}$ **do**
  Accept arm with estimated preferences
  Observe $Y^{(j)}_{A_j^{-1}(t)}(t)$ and update $\hat{\mu}^{(j)}_{A_j^{-1}(t)}, T^{(j)}_{A_j^{-1}(t)}$ if $\bar{A}_j^{-1}(t) = A_j^{-1}(t)$
end
Compute $\text{UCB}^{(j)}_n$ and $\text{LCB}^{(j)}_n$ for each $n \in [N]$
if $\exists \beta$ such that $\text{LCB}^{(j)}_{\beta_n} > \text{UCB}^{(j)}_{\beta_{n+1}}$ for any $n \in [N-1]$ **then**
  Preferences $= \beta$
else
  Preferences = arbitrary but fixed
end

`// Perform Gale-Shapley using the Preferences` $\beta^{(j)} = \beta = (\beta_1^{(j)}, \beta_2^{(j)}, \ldots, \beta_N^{(j)})$ `found`

Propose using β till acceptance
Initialize $s = 1$
for $t = t_{\text{exp}} + 1, \ldots, T$ **do**
  Accept player using the estimated preference $\beta^{(j)}$
  $s = s + 1$ if $\bar{A}_j^{-1}(t) == \emptyset$
end

1.2.2 `CA-ETC`

Algorithm 2: Base Algorithm: Expore-then-Gale-Shapley (view of arm a_j)

Input : Exploration rounds $2^l T_0$, Horizon $b^l T_0$

`// Learning Preferences`

for $t = \sum_{l'=1}^{l-1} 2^{l'} T_0 + 1, \ldots, \sum_{l'=1}^{l-1} 2^{l'} T_0 + 2^l T_0$ **do**
  Accept arm with estimated preferences
  Observe $Y^{(j)}_{A_j^{-1}(t)}(t)$ and update $\hat{\mu}^{(j)}_{A_j^{-1}(t)}, T^{(j)}_{A_j^{-1}(t)}$ if $\bar{A}_j^{-1}(t) = A_j^{-1}(t)$
end
Compute $\text{UCB}^{(j)}_n$ and $\text{LCB}^{(j)}_n$ for each $n \in [N]$
if $\exists \beta$ such that $\text{LCB}^{(j)}_{\beta_n} > \text{UCB}^{(j)}_{\beta_{n+1}}$ for any $n \in [N-1]$ **then**
  Preferences $= \beta$
else
  Preferences = arbitrary but fixed
end

`// Perform Gale-Shapley using the Preferences` $\beta^{(j)} = \beta = (\beta_1^{(j)}, \beta_2^{(j)}, \ldots, \beta_N^{(j)})$ `found`

Propose using β till acceptance
Initialize $s = 1$
for $t = 1, 2, \ldots, b^l T_0 - 2^l T_0$ **do**
  Accept player using the estimated preference $\beta^{(j)}$
  $s = s + 1$ if $\bar{A}_j^{-1}(t) == \emptyset$
end

Algorithm 3: Main Algorithm: Epoch-based `CA-ETC` (view of player a_j)

Input: Epoch length T_0, Parameter $\gamma \in (0,1)$ with $b = 2^{1/\gamma}$

for $l = 1, 2, \ldots$ **do**

  `Base Algorithm (Exploration rounds =` $2^l T_0$`, Horizon length =` $b^l T_0$`)`

end

1.2.3 Analysis

Theorem 1.5 (Regret of Algorithm 1)**.** *Suppose algorithm 1 is played for T iterations. Then, arm a_j incurs the regret of*

$$\overline{RA}_j(t) \leq \left(\frac{64K \log T}{\Delta^2} + K^2 + \frac{2NK\pi^2}{3} \right) \bar{\Delta}^{(j)}_{\max}.$$

Theorem 1.6 (Regret of Algorithm 3)**.** *Suppose `CA-ETC` is run with initial epoch length T_0 and input parameter $\gamma \in (0,1)$. Then, provided the initial epoch satisfies*

$$T_0 \gtrsim \left[\frac{32K \log T}{\Delta^2 (T-N)^\gamma} \right]^{\frac{1}{1-\gamma}},$$

the (arm-pessimal) regret for arm a_j is given by

$$\begin{aligned} \underline{RA}_j(t) \leq\ & T_0 \left(\frac{64K \log T}{T_0 (\Delta^{(j)})^2} + 4 \right)^{1/\gamma} \Delta^{(j)}_{\max} \\ & + 2T_0 \left(\frac{T-N}{T_0} + 1 \right)^{\gamma} \Delta^{(j)}_{\max} + K^2 \gamma \log \left(\frac{T-N}{T_0} + 1 \right) \Delta^{(j)}_{\max} + \frac{2NK\pi^2}{3} \Delta^{(j)}_{\max}. \end{aligned}$$

The proof of these follow the same steps as the proof for the players, and hence skipped.

References

[1] F. Kong and S. Li, *Player-optimal Stable Regret for Bandit Learning in Matching Markets*, pp. 1512–1522. [Online]. Available: https://epubs.siam.org/doi/abs/10.1137/1.9781611977554.ch55